VIETNAM NATIONAL UNIVERSITY – HCMC

UNIVERSITY OF SCIENCE

Faculty of Information and Technology

Software Engineering Department

Project proposal

Project name: Master Interview

Course: Introduction to Software Engineering

Class: 24C10

Students:

24127186 - Giáp Đăng Khoa

24127518 - Trần Ngọc Quân

24127136 - Nguyễn Thanh Trọng

24127137 - Nguyễn Thanh Trúc

Table of Contents

Project Proposal

Objectives

This document focus on the following topics:

- ✓ Completing the Project Proposal document with the following sections:
  - Preliminary Problem Statement
  - Proposed Solution
  - Development Plan
  - Human Resources & Costing Plan
- ✓ Understanding the Project Proposal document.
- ✓ This document will be used as input for AI tools to verify the quality of subsequent project artifacts.

All project artifacts must remain consistent and synchronized.

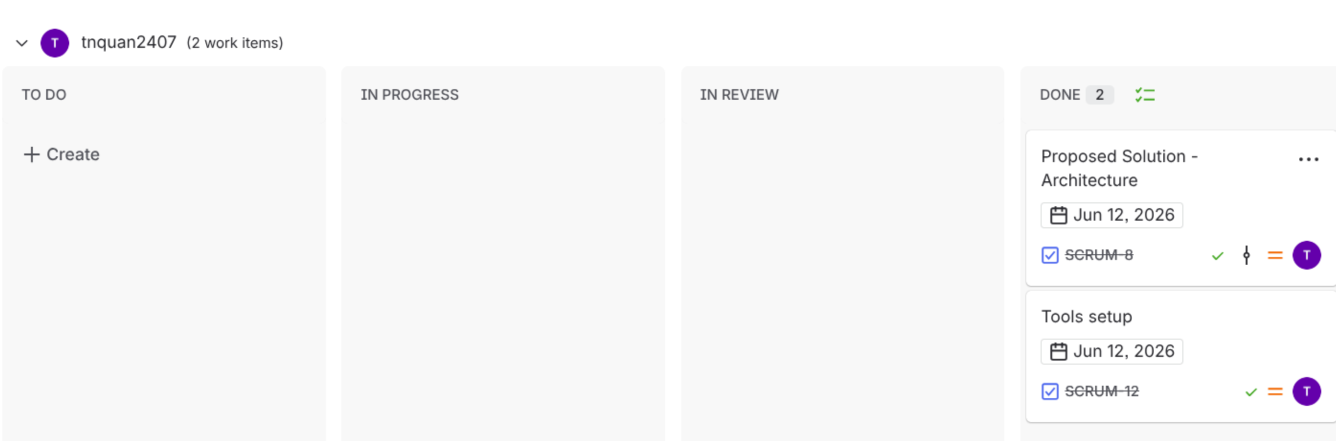
For example, if the project proposal is modified during this phase, a new version of the proposal must be documented.

By the conclusion of the project, all versions of every artifact must be submitted to demonstrate the evolution of your work.

1 Member Contribution Assessment

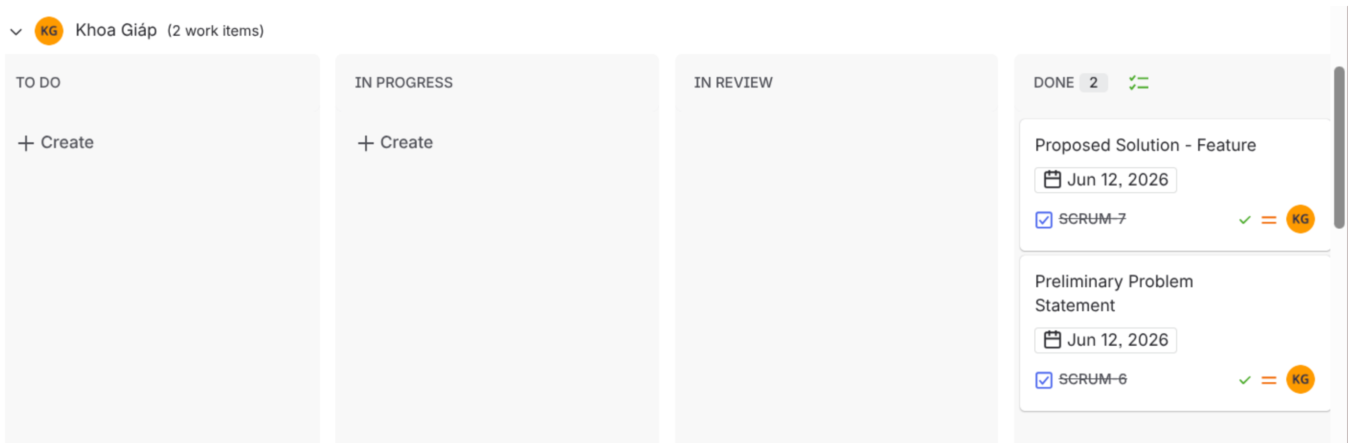
24127518 - Trần Ngọc Quân - Contribution (25%)

Task name	Summary
Write report: proposed solution - architecture	The architecture of the project is clearly stated.
Write report: Tools setup	The working environment has been set up and reported.



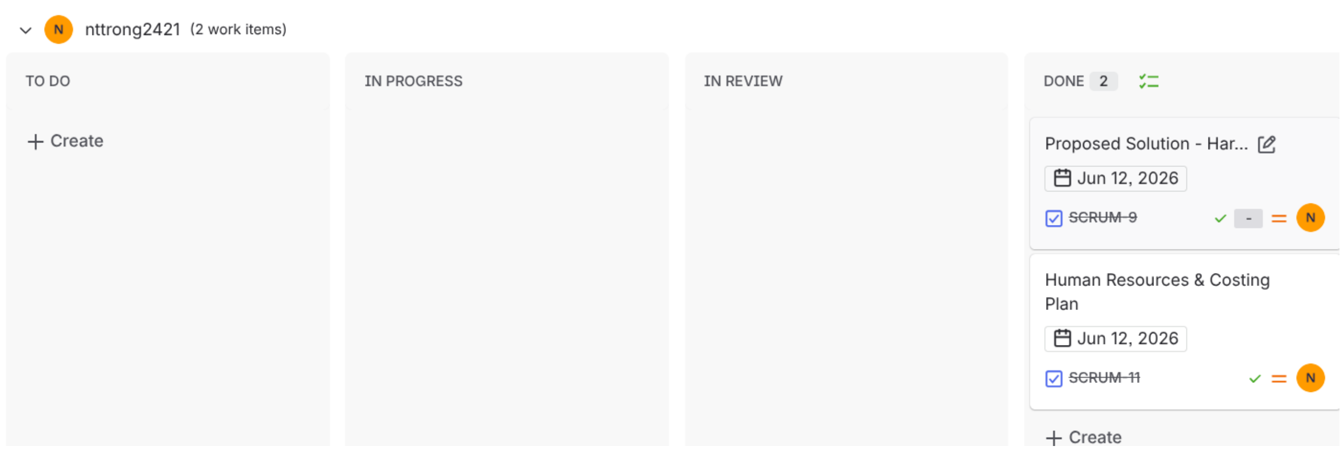
24127186 - Giáp Đăng Khoa- Contribution (25%)

Task name	Summary
Write report: proposed solution - feature	Every feature is based on user problem
Write report: Problem statement	Clearly state user story



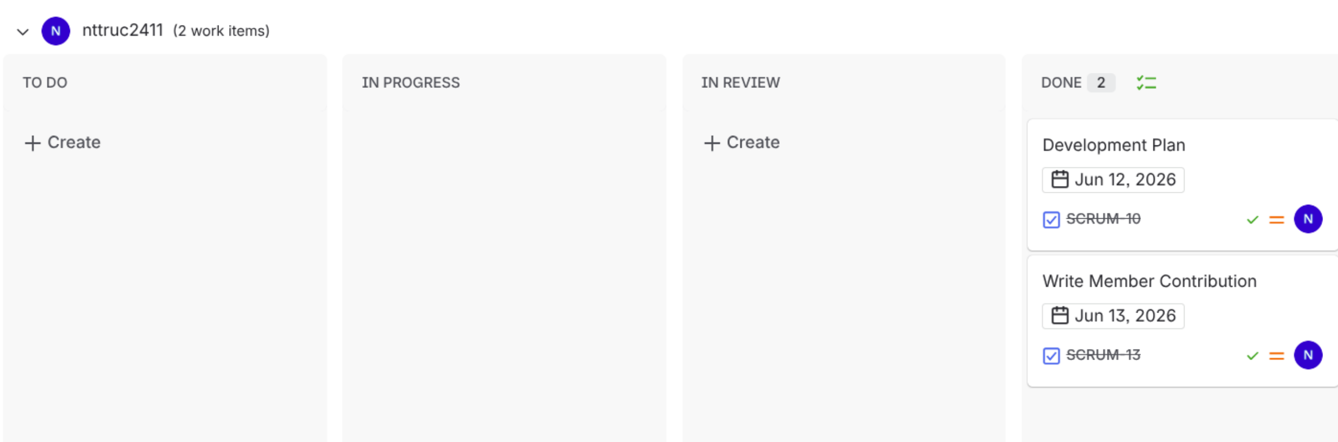
24127136 - Nguyễn Thanh Trọng - Contribution (25%)

Task name	Summary
Write report: proposed solution - hardware	Hardware based on the student PC statistic
Write report: Human resources & costing plan	Including time, money estimation



24127137- Nguyễn Thanh Trúc - Contribution (25%)

Task name	Summary
Write report: Member contribution	List out member contribution
Write report: Development plan	Make a feasible plan for the project



Each task must be assigned to and completed by only one student.

The sum of all student contributions is less than or equal to 100%.

The project tasks include report writing, self-training, and implementation tasks.

2 Preliminary Problem Statement

Please include the ID and full name of the student who authored, reviewed, or updated this section.

Written by: 24127186 - Giáp Đăng Khoa

Edited by:

Reviewed by: 24127518 - Trần Ngọc Quân

2.1 Business Description:

In the tech recruitment market, recent graduates or career switchers (**inexperienced group**) often face a barrier: they have foundational knowledge but lack practical skills and the psychological readiness required when facing recruiters. Contrastly, experienced industry experts (**Interviewers**) want to share their expertise, optimize their free time to generate income, and help candidates fully show their capabilities in real interviews - especially those who, due to a lack of confidence or interview experience, fail to demonstrate their true potential.

The **Mock Interview Platform (MasterInterview)** is developed and operates as a **web-based platform**, we provide through two core solutions: **Mock Interviews with Experts (Human-to-Human)**

The system operates based on two main user roles and an intelligent entity (AI Agent):

•**The Interviewee (Candidate):** The user needs to practice the interview. The basic process includes: uploading a resume (CV), selecting a specific job position, and choosing a preferred interview format.

•**The Interviewer (Expert):** Experienced engineers and managers from technology companies. They provide professional profiles, set available time slots, and determine the fee for each interview.

2.2 Revenue Model

The platform operates on a commission-based model:

- **For Human-to-Human bookings:** The system automatically deducts a **10% commission** from the total amount of each successfully completed transaction.

3 Proposed Solution

3.1 *Software*

3.1.1. Features

Please include the ID and full name of the student who authored, reviewed, or updated this section.
Written by: 24127186 - Giáp Đăng Khoa
Edited by:
Reviewed by: 24127518 - Trần Ngọc Quân

User Story Mapping & Requirements:

User Story	Features
As an Interviewee, I want to find and book experts based on my target job position so that I can practice relevant interview skills.	**Search & Booking System:** Filter experts by role, view profiles, and book available time slots.
As an Interviewee, I want my payment to be held securely until the interview is completed to ensure I receive the service I paid for.	**Escrow Payment Mechanism:** Funds are temporarily held by the system and released post-interview.
As an Interviewer, I want to set my own availability schedule and session fees so I can manage my time and earnings efficiently.	**Expert Profile Management:** Tools to configure availability slots and dynamic pricing.
As a user with a confirmed booking, I want the system to automatically provide a meeting link so I don't have to manually set it up.	**Third-party Meeting Integration:** Automated Zoom API link generation upon booking confirmation.
As an Interviewee, I want to rate the expert after the session so I can help other candidates make informed booking decisions.	**Post-Interview Rating System:** Feedback and rating submission that dynamically updates the expert's platform rank.
As a Candidate, I want my resume to be analyzed automatically so I can get instant, actionable feedback for specific job roles.	**AI CV Evaluation:** Automated resume parsing and scoring system.
As a user, I expect the AI Agent to respond to my	**Real-Time Processing:** AI interaction response

voice or text inputs with minimal latency during a live interview.	time should ideally be under 2-3 seconds to ensure natural dialogue flow.

The system is divided into two main processing:

a) Human Mock Interview:

- **Search & Booking:** Interviewees can look up the list of Experts based on job positions, view their profiles, availability slots, and ratings/reviews from previous users to proceed with booking a session.
- **Online Meeting Integration:** Once a booking is confirmed, the system connects with third-party services (via Zoom APIs) to generate a video meeting room and send the link access details to both parties.
- **Payment Mechanism:** To protect the interests of both sides, the Interviewee's payment is held in escrow by the system upon a successful booking. After the session concludes and is confirmed complete by both parties (or after a specified period without any problems), the system automatically releases the funds to the Interviewer (minus the 20% service fee).
- **Post-Interview Evaluation:** Interviewees provide feedback, rate the quality, and leave reviews regarding the Expert's interview performance to update the platform's **Interviewer rating system**.

b) AI Mock Interview:

- **CV Assessment:** The AI Agent integrates deep analysis capabilities to evaluate resume content, provide scoring, and offer tailored optimization suggestions for the targeted job position.

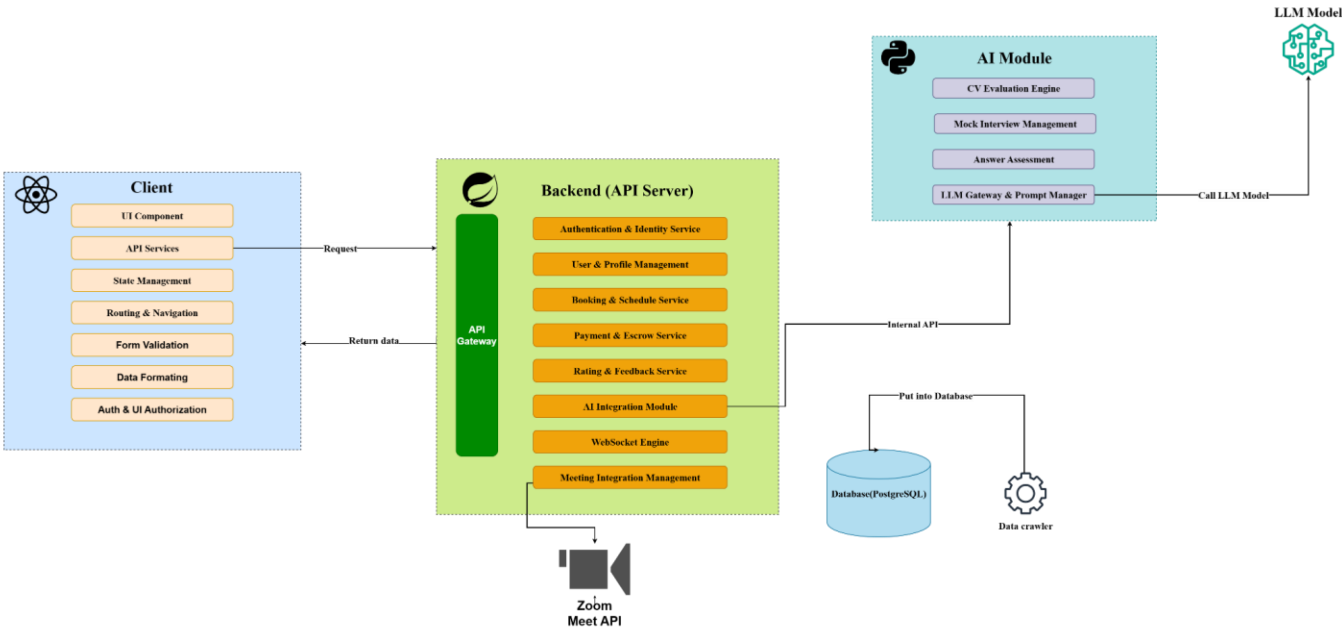
3.1.2. Software Architecture:

Please include the ID and full name of the student who authored, reviewed, or updated this section.
Written by: 24127518 - Trần Ngọc Quân
Edited by:
Reviewed by: 24127136 - Nguyễn Thanh Trọng

The software system is built upon a Client-Server architecture, incorporating a Service-Oriented approach to decouple the core operations from the AI processing.

- **Client (Frontend):** Developed using React, this layer is responsible for the User Interface (UI), state management, and communicating with the backend services via REST APIs (and WebSockets for real-time interactions).
- **Backend (API Server):** Built with Spring Boot, it acts as the core engine that handles central business logic, user management, security, and Database interactions.

- **AI Module:** A standalone service dedicated to processing AI-heavy tasks, such as CV evaluation and managing the Mock Interview flow, which interacts with external LLMs and communicates with the Backend via internal APIs.
- **Data Crawler**: A background process responsible for searching and extracting interview questions from external sources to populate the system's database.



3.2 Hardware

Please include the ID and full name of the student who authored, reviewed, or updated this section.

Written by: 24127136 - Nguyễn Thanh Trọng

Edited by:

Reviewed by: 24127186 - Giáp Đăng Khoa

The student team should describe the hardware and equipment requirements necessary for the software to operate.

Operating System: Windows 11

Processor (CPU): Intel(R) Core(™) i7-14700HX

Graphics (GPU): NVIDIA GeForce RTX 4060 Laptop GPU

Memory (RAM): 16GB

4 Development Plan

Please include the ID and full name of the student who authored, reviewed, or updated this section.

Written by: 24127137 - Nguyễn Thanh Trúc

Edited by:

Reviewed by: 24127186 - Giáp Đăng Khoa

The student team should present a detailed software development plan, along with the specific deliverables for each phase.

4.1 *Requirements Analysis*

- Timeline: 16/6 - 28/6
- Objectives:
  - Document functional, non-functional requirements
  - Identify project scope, technical constraints, feasibility
- Deliverables:
  - Functional requirements
  - Non-functional requirements: performance, safety, security, usability, scalability
  - Use case diagram: actors, use cases, environment
  - Use case details

4.2 *Software Design*

- Timeline: 29/6 - 13/7
- Objectives: Blueprint for future implementation
  - Plan system architecture
  - Design database schema
  - Design user interface and user experience.
- Deliverables:
  - System architecture: Diagrams for tech stack, data flow

    - Database schema
    - UI/UX wireframe

4.3 Implementation

- Timeline: 14/7 - 20/8
- Objectives: MVP - Minimum viable product, pass unit tests
- Deliverables: a website that cover core features:
    - Expert profile management
    - Search and booking system
    - Third-party meeting
    - AI mock interview in text
    - AI CV evaluation

4.4 Testing

- Timeline: 21/8 - 24/8
- Objectives: pass integration test cases, fix existing bug
- Deliverables:
    - Test plan
    - Test cases
    - Test result

4.5 Deployment and Maintenance

- Timeline: 24/8
- Objectives: Viable product, documentation
- Deliverables:
    - Error-free website that cover core features
    - Documentations: Technical document, user manual

5 Human Resources & Costing Plan

Please include the ID and full name of the student who authored, reviewed, or updated this section.

Written by: Nguyễn Thanh Trọng

Edited by:

Reviewed by:

The student team should present the personnel structure for the entire project or each development phase and list the anticipated costs.

To optimize the productivity for a group of 4 members within a short period of time, the human resources are allocated following a Scrum model. This ensures all members actively contribute to key engineering phases according to the academic milestones.

- Core Roles Distribution

Student's ID	Student's Name	Role
24127518	Trần Ngọc Quân	Back-end Developer
24127186	Giáp Đăng Khoa	UI/UX Designer + Prompt Engineer
24127137	Nguyễn Thanh Trúc	Data crawling + AI Integration Developer
24127136	Nguyễn Thanh Trọng	Front-end Developer + UX/UI Designer

- Resource Allocation:

Phase	Timeline	Member	Focus & Responsibility Assignment
		Quân	Designing software architecture

1. Project Proposal	10/06/2026 - 14/06/2026	Khoa	Stating problem; Giving solutions
		Trúc	Proposing development plan for each phase
		Trọng	Anticipating the costing; Allocating human resource
2. Requirement Analysis	16/06/2026 - 28/06/2026	Quân	Designing use case diagram of the system; Testing use case
		Khoa	Stating problem; Defining functional & Non-functional Requirements; Illustrating use case
		Trúc	Scraping and cleaning data; Calling API from third party
		Trọng	Designing UI/UX, prototype on Figma
3. System Design	29/06/2026 - 13/07/2026	Quân	Presenting EER, architecture and class diagrams
		Khoa	Giving explanations and analysis on data tables
		Trúc	Presenting data diagram
		Trọng	Presenting screen diagram; Listing important screens and the sequence of events
4. Testing & Deployment	21/08/2026 - 24/08/2026	Quân	Comprehensive cross-testing and production deployment, jointly executed by all 4 members
		Khoa	
		Trúc	
		Trọng	

- Token cost:

Model	Input Tokens ($/MTok)	Amount	Output Tokens ($/MTok)	Amount	Total ($/MTok)
Gemini 3.5 Flash	1.5	2	9.0	1	12
Gemini 3.1 Flash Live Preview	Text: 0.75 Audio: 3.0 or 0.005/min	2	Text: 4.5 Audio: 12.0 or 0.018/min	1	Text: 6 Audio: 18

- Time cost:

  Labor time breakdown:

  - Phase 1: Approximately 2hours/day/member, which is 40 person-hours in total.
  - Phase 2: Approximately 3hours/day/member, which is 168 person-hours in total.
  - Phase 3: Approximately 3hours/day/member, which is 168 person-hours in total.
  - Phase 4: Approximately 3hours/day/member, which is 168 person-hours in total.

6 Tools setup

Please include the ID and full name of the student who authored, reviewed, or updated this section.

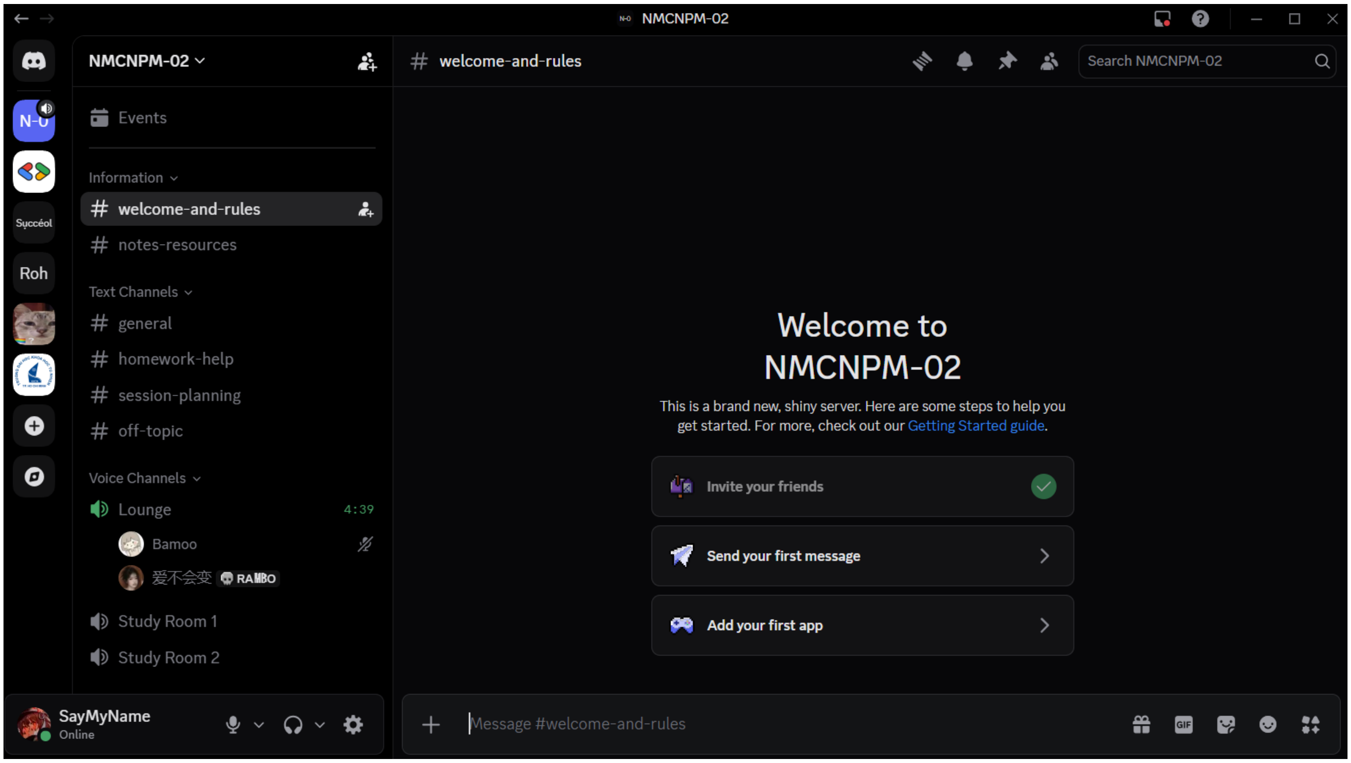
Written by: 24127518 - Trần Ngọc Quân

Edited by:

Reviewed by: 24127136 - Nguyễn Thanh Trọng

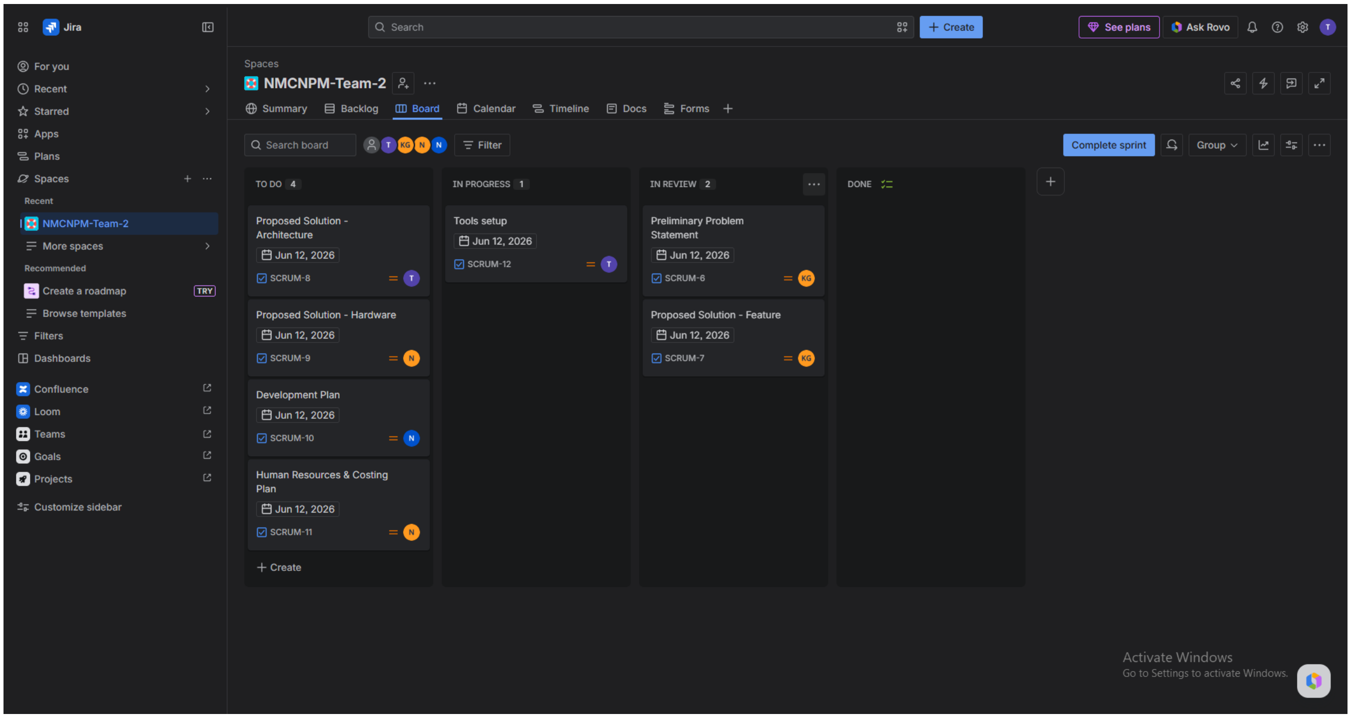
To support effective collaboration and project management throughout the development process, our team adopted the tools required by the course. These tools were used for communication, task management, source code control, documentation, and assignment submission. The following sections describe the purpose and usage of each tool in our project workflow:

- **Moodle**: Used for posting and submitting assignments.
- **Discord**: The primary communication platform for discussions and interactions among team members.



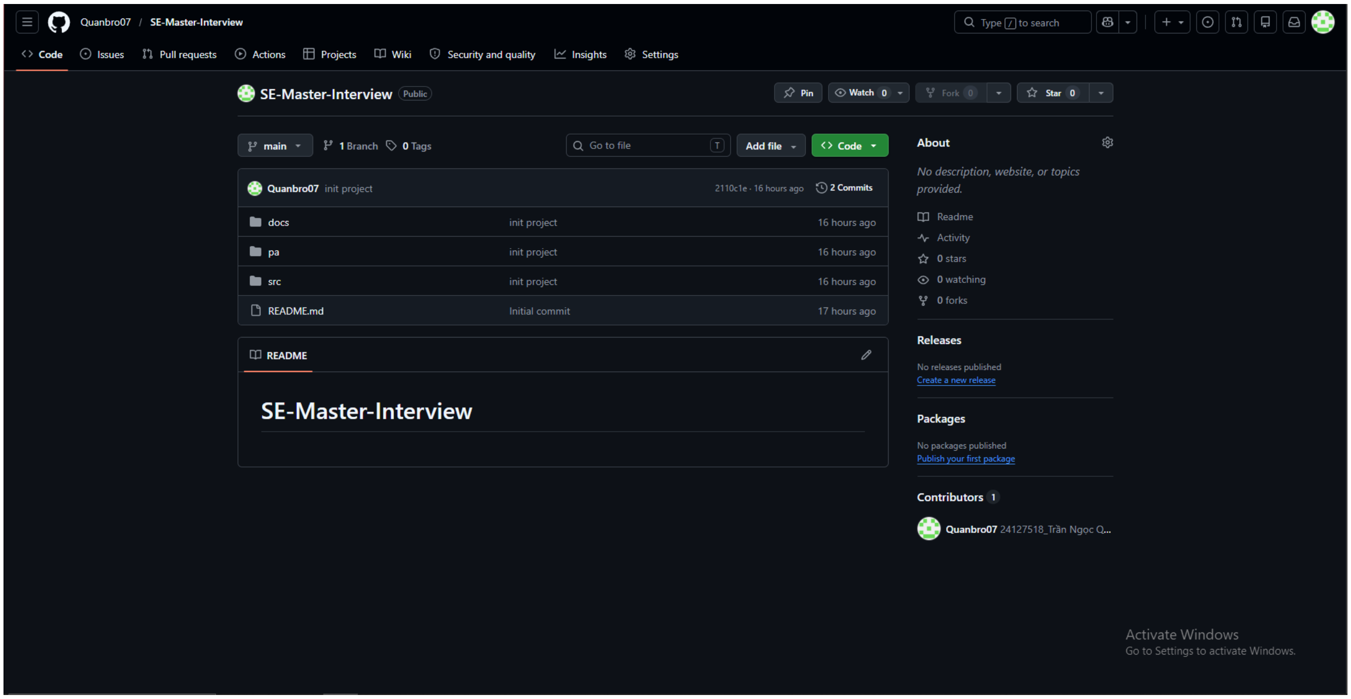
[Figure 1. Discord]

- **JIRA**: Used for task assignment, sprint planning, schedule tracking, and project timeline management. It is integrated with GitHub through webhooks, allowing commits and pull requests to be automatically linked to Jira issues.



[Figure 2. Jira Scrum Board]

- **GitHub**: Used for source code management, version control, and documentation. The repository follows folder structure described below:
  - /src: Used to store source code.
  - /docs: Used to store documentation, which has these subfolders:
    - management: Storing planning documents, reports (weekly report, project status report, etc.).
    - requirements: Storing all requirements, including vision documents and use cases.
    - analysis and design: Storing all analysis and design related documents, including software architecture documents, UML models, and UI designs.
    - test: Storing all test documents such as test plans, test cases, and test reports.
  - /pa: Including subfolders to store submissions. Each subfolder contains one PA submission.



[Figure 3. GitHub Repository Structure]

7 AI Usage Declaration

Based on the provided AI Usage Guideline document, teams must submit their AI usage declaration here. Students are encouraged to leverage AI tools creatively and effectively in this project.

Gemini. Gemini 3.1, Google, gemini.google.com, accessed [07:31 on June 10, 2026], prompt: "How to integrate and trigger JIRA webhooks via GitHub", used for the [system integration / CI/CD setup]; AI provided the configuration steps and payload structure, student implemented the connection, verified the integration in practice, and documented the process in their own words.

Gemini. Gemini, Google, gemini.google.com, accessed 16:18 on June 11, 2026, prompt: "Trong các project software người ta thường dùng tool gì để theo dõi deadline cũng như tiến trình viết theo format trên dùng cho chapter 0, tìm kiếm công cụ cho theo dõi công việc", used for the project management tools and task tracking methodology section in Chapter 0; AI provided a comprehensive list of industry-standard software project management tools (such as Jira, Trello, and Linear), student rewrote in academic language, selected the most suitable tracking tool for the team, and established the project development timeline.

8 Presentation

The link to the presentation is provided below

https://youtu.be/M0yle9vRROM

9 Reflective Report

Most Helpful for Implementation: Sections **2 (Preliminary Problem Statement)**, **3 (Proposed Solution)**, and **4 (Development Plan)** are highly practical as they serve as the technical blueprint for the project.

- **Reasoning & Example:** Sections 2 and 3 define the core logic and system architecture. For instance, clearly outlining the solution allows the team to properly design the database schema and define backend APIs before writing the actual code for the frontend. Section 4 provides a concrete roadmap, making it easy to divide coding tasks and track progress week by week.

Unnecessary / Less Practical: Sections **5 Costing Plan**, are mostly administrative and lack practical value for a small-scale student project.

- **Reasoning & Example:** Section 5 is entirely theoretical. Since we are a student team building the application without a real financial budget, calculating hourly wages or projecting server costs does not impact the actual development process. Sections 6 and 7 are formal declarations that do not directly contribute to coding or system optimization.

10 References

Krüger, Gerhard. “How to Write a Software Requirements Specification (SRS) Document.” https://www.perforce.com/blog/alm/how-write-software-requirements-specification-srs-document. Accessed 2026.